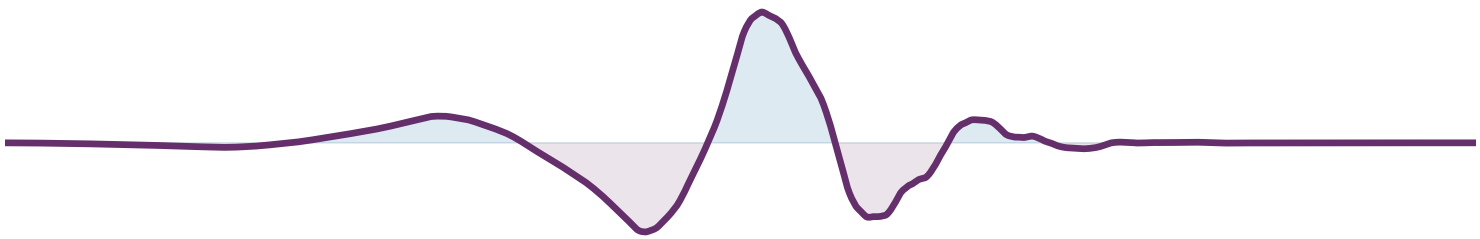
ECE 251C Lecture 13

Wavelets and Sparsity

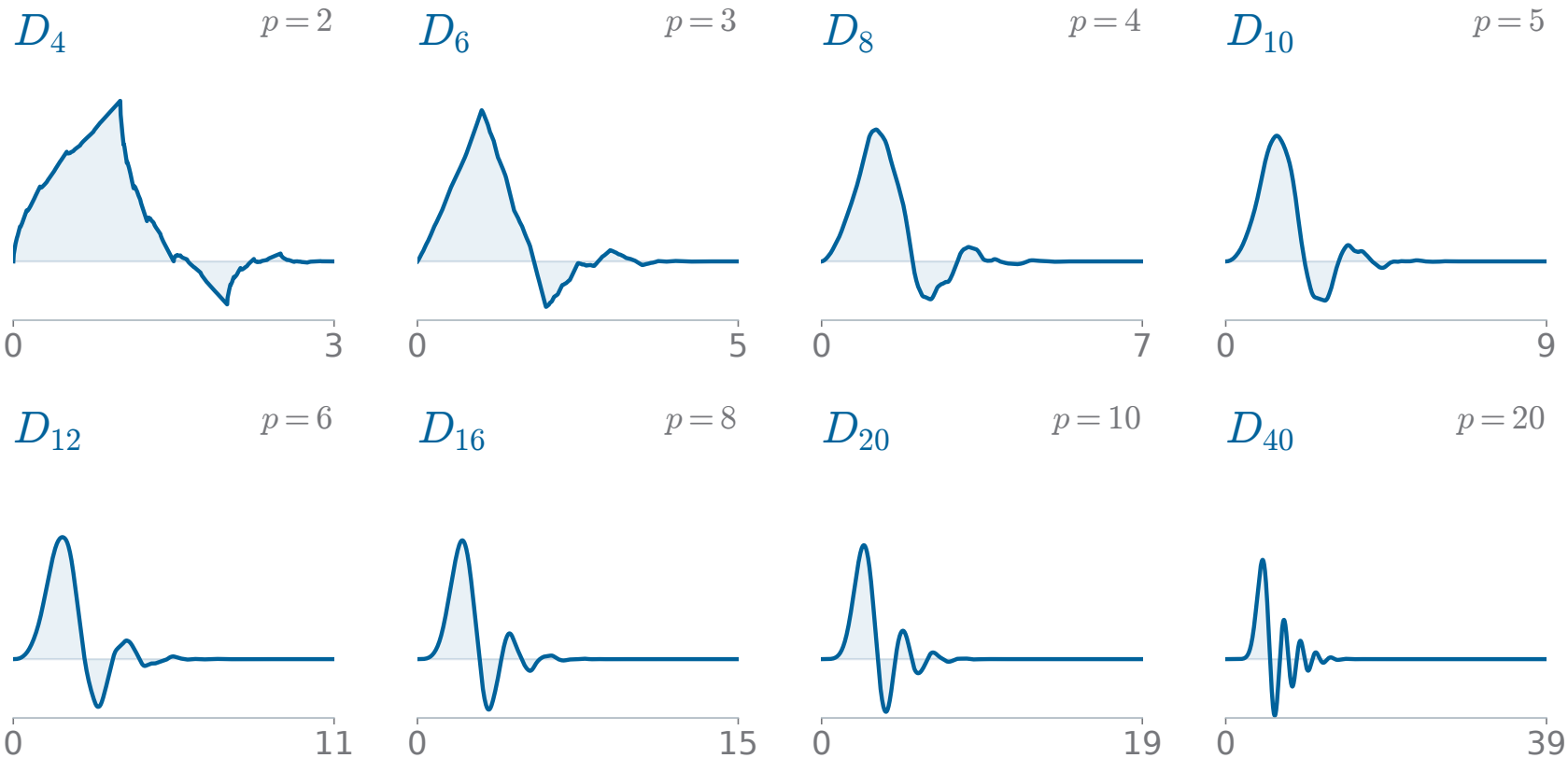
Rahul Parhi

Electrical and Computer Engineering
University of California, San Diego



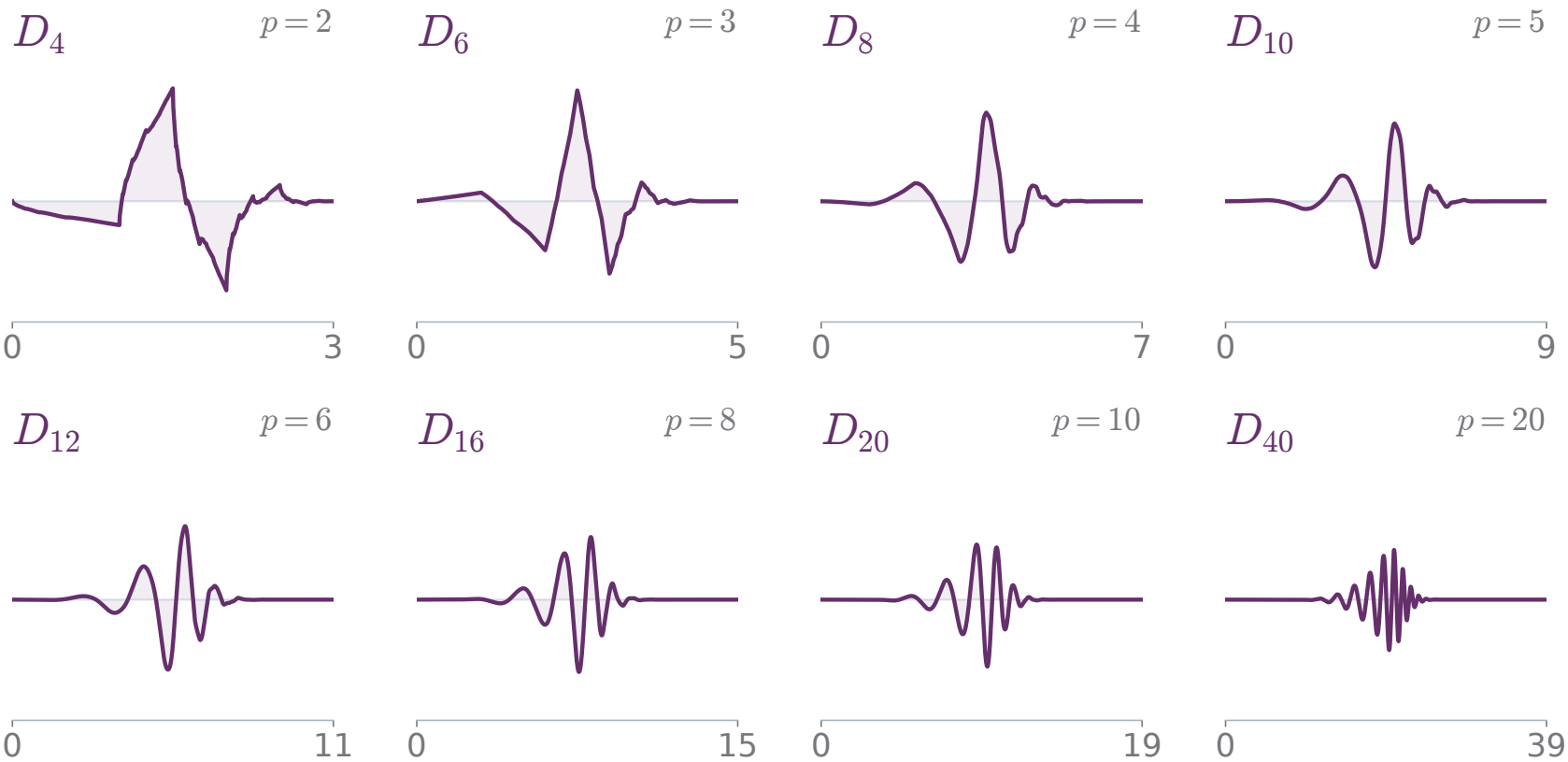
Daubechies Scaling Functions

For D_{2p}, the **associated wavelet** has p vanishing moments, and the filters have $2p$ **taps**.



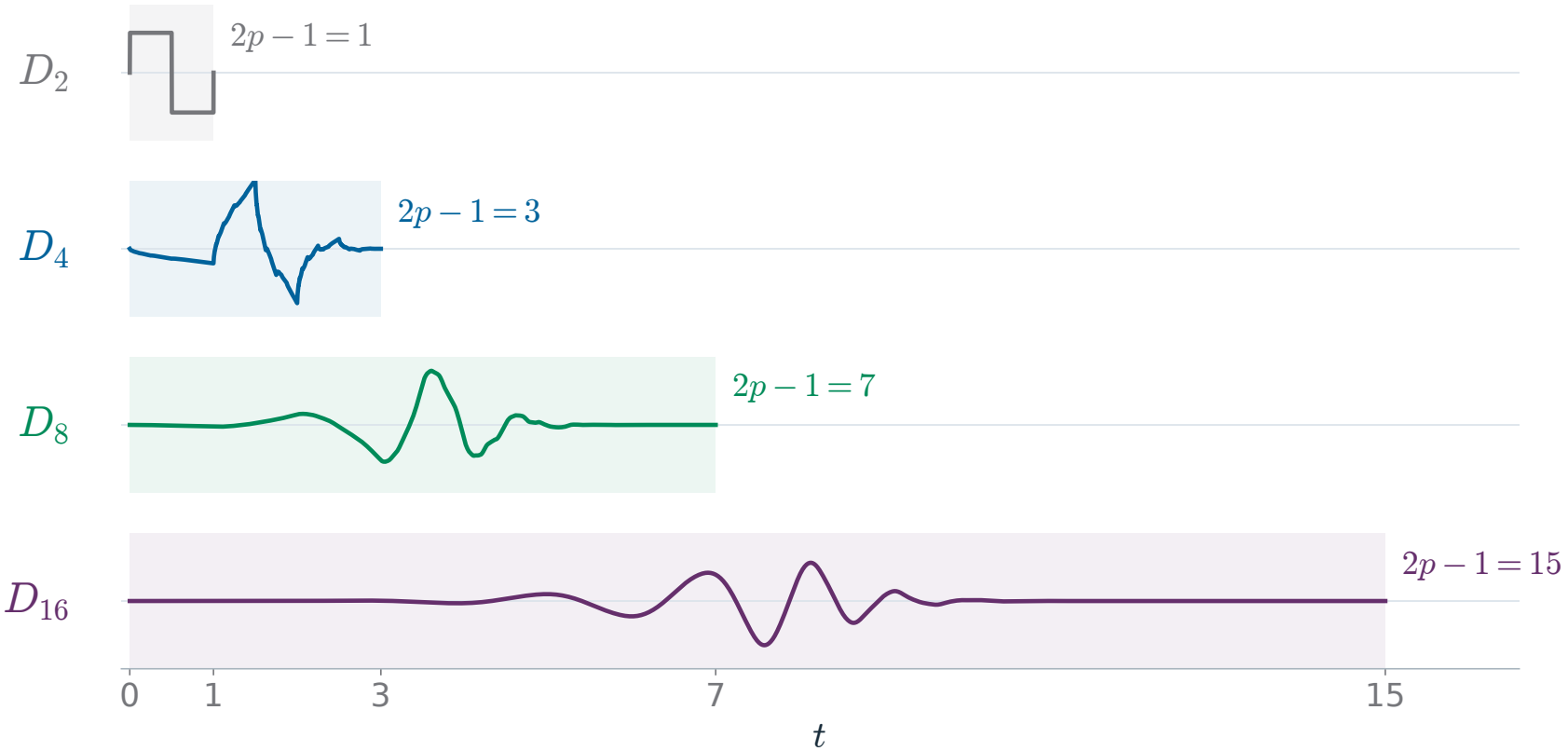
Daubechies Wavelets

The corresponding wavelets satisfy $\int_{\mathbb{R}} t^m \psi(t)\,\mathrm{d}t = 0,\ m = 0, \ldots, p-1.$



Vanishing Moments and Support

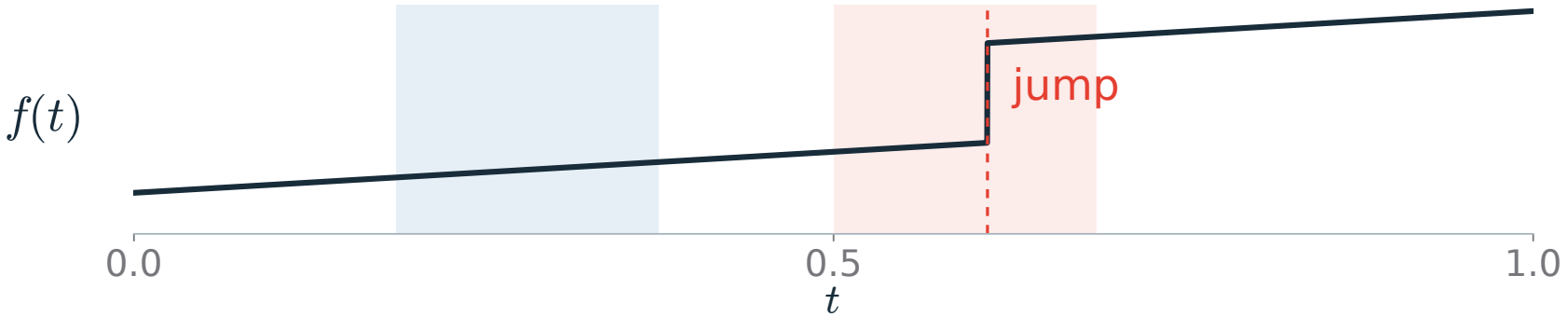
The support grows with p.



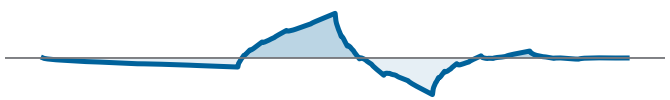
p vanishing moments $\quad$ $2p$ taps $\quad$ support length $2p-1$

Vanishing Moments Give Local Cancellation

For D_4, the wavelet is orthogonal to every affine function.

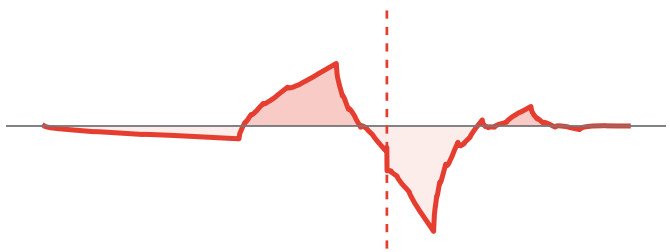


Support inside an affine piece



$$\int f(t)\,\psi_{i,n}(t)\,\mathrm{d}t = 0$$

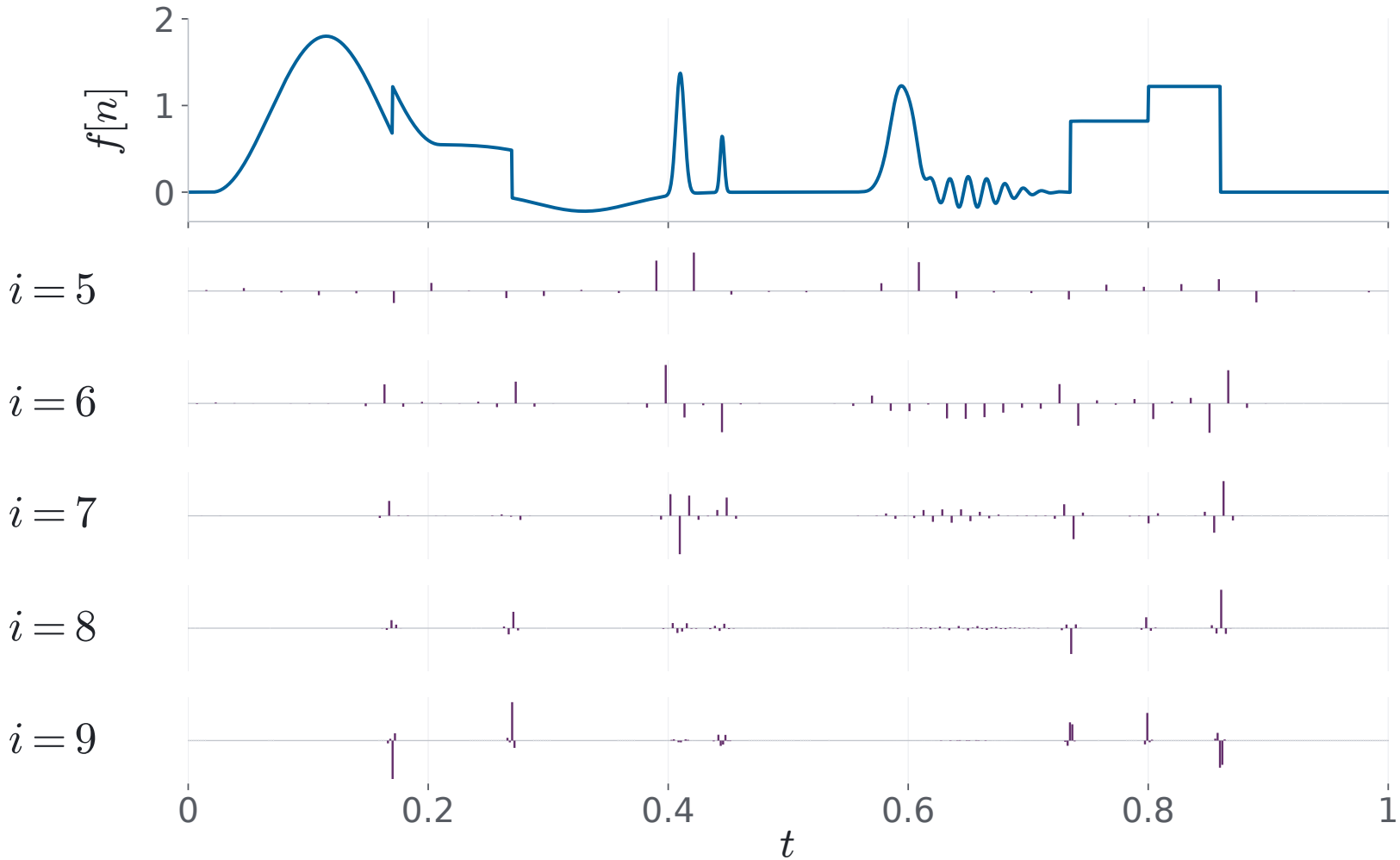
Support crosses the jump



$$\int f(t)\,\psi_{i,n}(t)\,\mathrm{d}t \neq 0$$

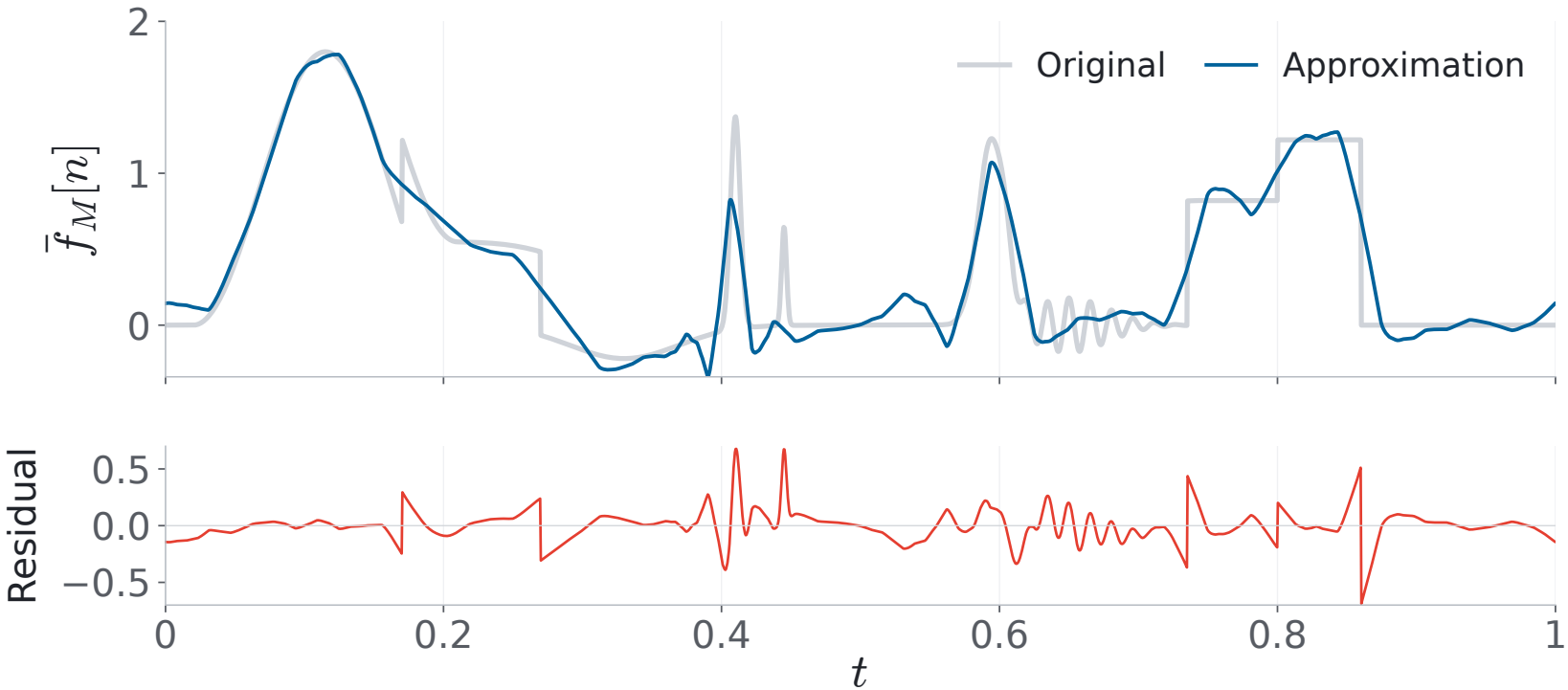
If f agrees with a polynomial of degree $< p$ on the **entire wavelet support**, then $\langle f, \psi_{i,n} \rangle = 0$.

A Sparse Wavelet Representation



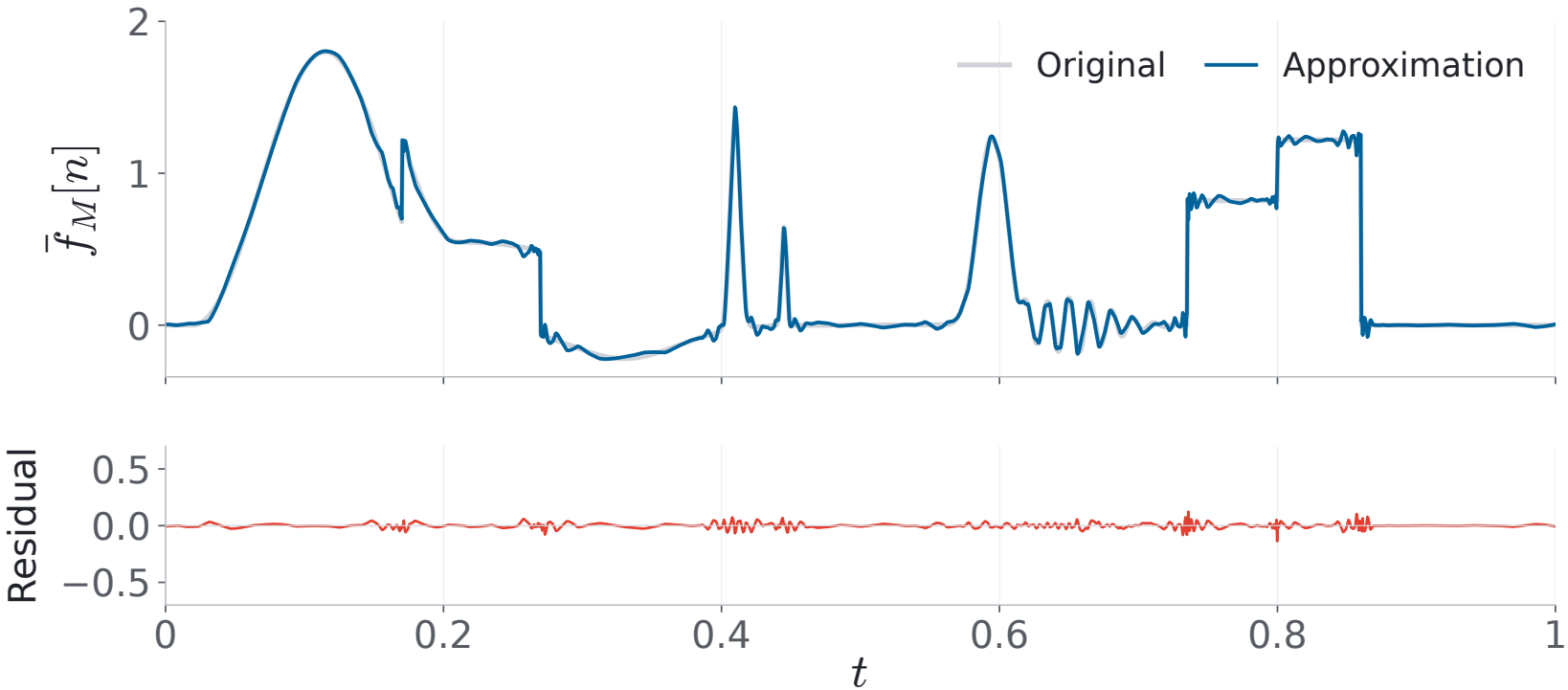
Reconstruction From a Few Coefficients

20 of 2048 coefficients $\approx 1\%$ retained 19.92% **relative error**



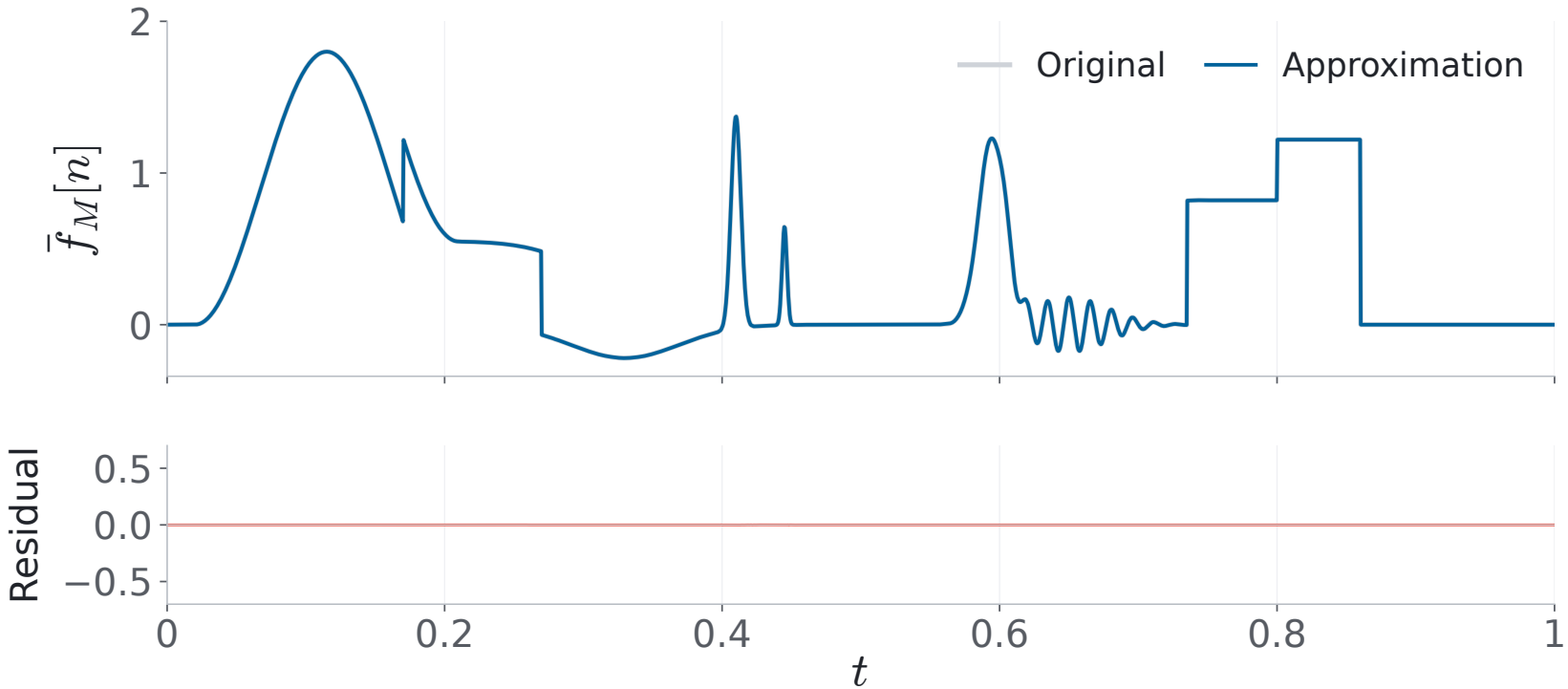
Reconstruction From a Few Coefficients

102 of 2048 coefficients $\approx 5\%$ retained 2.82% **relative error**



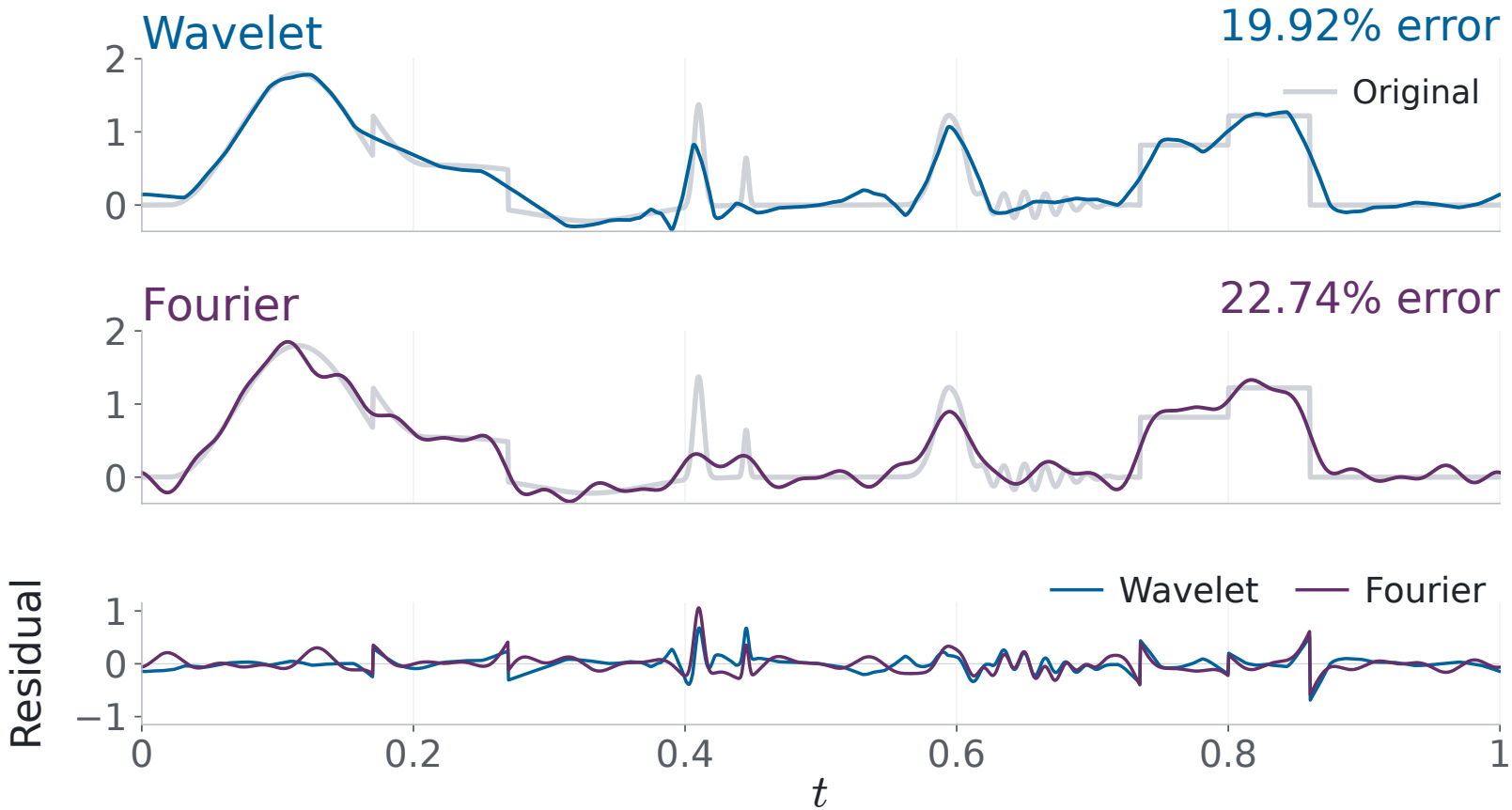
Reconstruction From a Few Coefficients

307 of 2048 coefficients $\approx 15\%$ retained 0.040% **relative error**



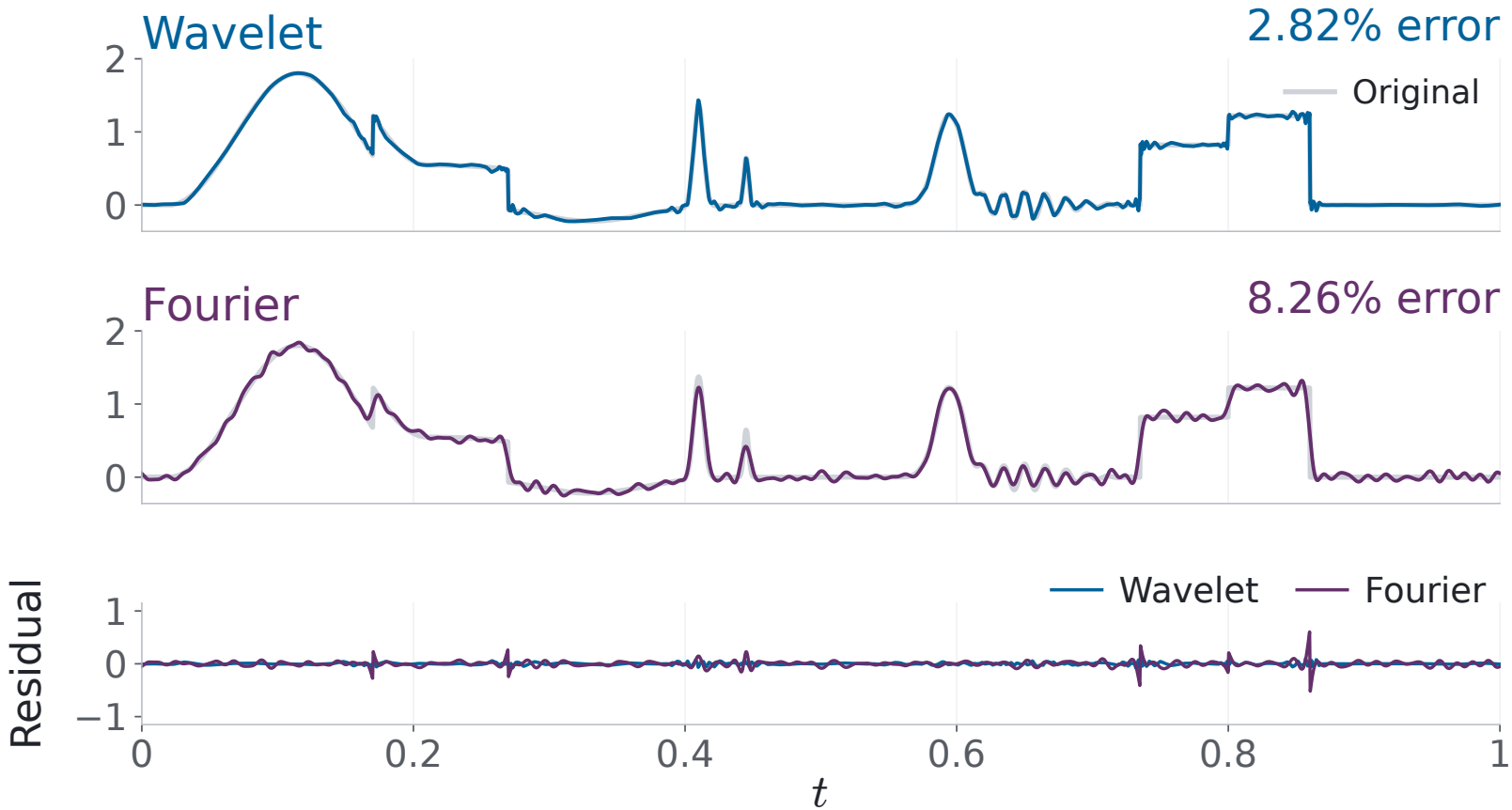
Wavelet and Fourier Thresholding

20 coefficients $\approx 1\%$ retained



Wavelet and Fourier Thresholding

102 coefficients $\approx 5\%$ retained



Wavelet and Fourier Thresholding

307 coefficients $\approx 15\%$ retained

